The Very Standard Tricks in Combinatorics

Version 1.4.1

15-7-2023

Contents

1 Counting

One kind of combinatorics problems is to count the number of certain things. We always need to count something when solving various problems. So it is important to know different counting methods.

1.1 Basic Counting

Problem 1.1.

(a) Find the number of 2-digit odd numbers whose sum of digits is a multiple of 5.

(b) Find the number of 3-digit numbers whose sum of digits is a multiple of 12.

Solution.

(a) The simplest way to count things is to list out all possible cases and count. Such a method is called the method of exhaustion. For this problem, the possible 2-digit numbers are $19, 23, 37, 41, 55, 69, 73, 87, 91$. So there are 9 such numbers. □

(b) We try to use the method of exhaustion. To count systematically, we divide into simpler cases. For example, we first count those numbers whose sum of digits is exactly 12. The numbers are

$$129, 138, \dots, 192, 219, 228, \dots, 291, 309, 318, \dots, 390, \dots, 903, 912, 921, 930.$$

There are $8 + 9 + 10 + 9 + 8 + 7 + 6 + 5 + 4 = 66$ numbers (counted according to the hundreds digit).

Next, we count those numbers whose sum of digits is exactly 24. We get the 10 numbers

$$699, 789, 798, 879, 888, 897, 969, 978, 987, 996.$$

Since the sum of digits of a 3-digit number cannot exceed 27, there are no other choices. In total, there are $66 + 10 = 76$ numbers whose sum of digits is a multiple of 12.

There may be other ways to count. For example, when the sum of digits is 24, we may further consider subcases according to the choices of the digits. Note that the digits can only be $6, 9, 9$; $7, 8, 9$ or $8, 8, 8$. Then we may count the number of ways to rearrange these digits and get $3 + 6 + 1 = 10$ as above. By using other classifications, one may get better (meaning faster) counting methods. □

Although we can list out the things in better ways to simplify the counting jobs, the method of exhaustion is still troublesome when there are more cases to consider.

Obviously, this method is also not reliable as one will probably make careless mistakes in counting (some cases are ignored or some are counted twice). The only advantage is that we do not need to think of how to count. Sometimes we may be able to count by listing during the time for considering other 'better' methods! And sometimes there could be no other methods except listing.

Problem 1.2.

(a) Find the number of 5-digit numbers all of whose digits are even.

(b) Find the number of 5-digit numbers whose sum of digits is a multiple of 5.

Solution.

(a) This time we shall not list out all possible numbers. We would like to have some methods which work for any number of digits. Indeed, one useful way of counting is to use the multiplication principle. Note that there are 4 choices for the first digit $(2, 4, 6, 8)$ and 5 choices for each of the remaining digits $(0, 2, 4, 6, 8)$. As the choices for these digits are independent, we deduce that there are $4 \times 5^4 = 2500$ desired numbers. In general, there are $4 \times 5^{n-1}$ n-digit numbers all of whose digits are even. □

(b) This is quite similar to problem 1.1(a). Instead of the method of exhaustion, we use a similar consideration as the first part. Firstly, there are 9 choices for the first digit. Next, we have 10 choices for each of the thousands, hundreds and tens digits. But how about the unit digit? We cannot choose any digit we like due to the constraint. It seems that the choice of the unit digit k depends on the other digits.

After the first 4 digits are chosen, the unit digit k should satisfy $k \equiv a \pmod 5$ for some fixed a (which is the negative of the sum of the first 4 digits). No matter what the value of a is, there are exactly 2 choices for k since $0 \leqslant k \leqslant 9$. This number '2' now becomes independent of the first 4 digits (although the deduction of this number comes from the other digits). So we can still apply the multiplication principle. This shows there are $9 \times 10^3 \times 2 = 18000$ such numbers.

In general, there are $18 \times 10^{n-2}$ n-digit numbers whose sum of digits is a multiple of 5 if $n \geqslant 2$. But this solution does not work if we try to count those numbers whose sum of digits is 8, for example. □

Many counting problems can be rephrased in terms of permutation and combination. In those cases, we can use their formulae to count directly to save a lot of time (the formulae can be proved by using the multiplication principle). In particular, the binomial coefficient $\binom{n}{k} = \dfrac{n(n-1)\cdots(n-k+1)}{k!}$ is much more useful.

Problem 1.3. In how many ways can we use the digits $0, 1, 1, 1, 2, 2, 3, 4$ to form an 8-digit number?

Solution. We first assume all digits are distinct. Since the first digit cannot be 0, there are 7 choices for the first digit. Afterwards, there are 7! ways to permute the remaining digits. By the multiplication principle, we obtain $7 \cdot 7! = 35280$ numbers. However, some numbers are the same because there are 3 identical 1's and 2 identical 2's. Since there are $3! = 6$ ways to permute the 1's and $2! = 2$ ways to permute the 2's, each number is counted $6 \times 2 = 12$ times. Thus, the correct answer is $\dfrac{35280}{12} = 2940$. □

Problem 1.4.

(a) Find the number of nonnegative integer solutions to $w + x + y + z = 10$.

(b) Find the number of positive integer solutions to $w + x + y + z = 10$.

(c) Find the number of integer solutions to $w + x + y + z = 10$ such that $w \geqslant 3$, $x \geqslant 2$, $y \geqslant 1$ and $z \geqslant 0$.

Solution.

(a) This is a rather standard counting example, and we shall present a standard solution here. Suppose we have 10 identical objects (denoted by circles o). We try to separate the objects into 4 parts by 3 commas. For example, oo,ooo,oo,ooo is a possible way, which divides the 10 objects into 4 parts of $2, 3, 2, 3$ each. This corresponds to the solution $(w, x, y, z) = (2, 3, 2, 3)$. Note that this correspondence is one-to-one (meaning that each configuration corresponds to exactly one solution). But it is easy to count the number of permutations of 10 objects and 3 commas. Indeed, this is the same as putting 3 commas in $10 + 3 = 13$ positions, since the remaining spaces would be occupied by the objects. There are $\dbinom{13}{3} = \dfrac{13 \times 12 \times 11}{3 \times 2 \times 1} = 286$ ways to do so.

In general, there are $\dbinom{k+n-1}{n-1}$ nonnegative integer solutions to $x_1 + x_2 + \cdots + x_n = k$ for nonnegative integer k. □

(b) Again, we have 10 objects oooooooooo, and we need to separate these objects into 4 parts so that each part has at least one object. This time we can put the 3 commas into the 9 spaces between the objects. Note that no 2 commas should be put into the same space. Therefore, the number of ways to do this task is exactly $\dbinom{9}{3} = \dfrac{9 \times 8 \times 7}{3 \times 2 \times 1} = 84$.

In general, there are $\dbinom{k-1}{n-1}$ positive integer solutions to $x_1 + x_2 + \cdots + x_n = k$ for integer $k \geqslant n$. □

(c) Rather than transforming the question to the above counting questions directly, we first do a little trick on the equation, so that the problem is reduced to one of the above situations. Indeed, we define $a = w - 2$, $b = x - 1$, $c = y$ and $d = z + 1$. Then the conditions become $w, x, y, z \in \mathbb{N}$, and the equation becomes $a + b + c + d = w + x + y + z - 2 = 8$. By the result stated in part (b), we know that there are $\binom{7}{3} = 35$ solutions in total. $\square$

Problem 1.5. In an 8×8 chessboard, a chess piece is placed in the bottom left corner cell. Each time we can only move the chess piece rightwards or upwards to an adjacent cell. In how many ways can we move the chess piece to the top right corner cell?

Solution. This is another standard problem that can be transformed to a combination problem. Note that no matter how the chess piece moves, it must eventually move rightwards for 7 times and move upwards for 7 times. Therefore, we can represent its route by 7 letters R and 7 letters U, where R and U stands for the rightward direction and the upward direction respectively. Conversely, each such permutation of 7 copies of R's and 7 copies of U's corresponds to a unique route. Therefore, it suffices to count the number of such permutations. This is the same as putting 7 copies of R's in 14 positions. Thus, the answer is $\binom{14}{7} = 3432$. $\square$

Problem 1.6. How many ways are there to choose 4 integers $1 \leqslant w < x < y < z \leqslant 12$ such that no 2 of them are consecutive?

Solution. If we just need to choose 4 integers from 1 to 12, then there are obviously $\binom{12}{4}$ ways. The constraint does not allow us to choose 2 integers which differ by 1. This condition is rather similar to the constraint we came across in problem 1.4(c). So we first make the following substitution. Let $a = w$, $b = x - 1$, $c = y - 2$ and $d = z - 3$ so that $1 \leqslant a < b < c < d \leqslant 9$ (for example, as w, x are not consecutive, we have $w < x - 1$, and hence $a < b$). Again, this correspondence is one-to-one. Therefore, there are $\binom{9}{4} = 126$ choices in total. $\square$

Problem 1.7. How many different combinations are there by throwing 5 dices together? (Note that the result of each dice can be $1, 2, 3, 4, 5, 6$.)

Solution. This is different from the usual combination $\binom{6}{5}$ since each number can appear more than once. For example, a possible combination is $1, 1, 2, 2, 6$. This kind of problems is essentially the same as problem 1.4(a). Indeed, suppose there are $a_1, a_2, \ldots, a_6$ dices resulting in the numbers $1, 2, 3, 4, 5, 6$ respectively. Then we are counting the number of nonnegative integer solutions to $a_1 + a_2 + \cdots + a_6 = 5$. By the result stated in problem 1.4(a), the answer is $\binom{5+6-1}{6-1} = \binom{10}{5} = 252$. $\square$

Sometimes when we count according to cases, some situations may belong to more than one case. Therefore, we cannot just apply the additional principle directly. If we do so, we have to subtract those extra cases in order not to overcount. A precise result is given by the following.

Theorem 1.1. (Inclusion-exclusion principle(容斥原理)) Let $S_1, S_2, \dots, S_n$ be any finite sets. Then

$$\begin{aligned}&|S_1 \cup S_2 \cup \cdots \cup S_n| \\ =& \sum_{i=1}^{n} |S_i| - \sum_{i<j} |S_i \cap S_j| + \sum_{i<j<k} |S_i \cap S_j \cap S_k| - \cdots + (-1)^{n+1} |S_1 \cap S_2 \cap \cdots \cap S_n|.\end{aligned}$$

This is theorem 2.4 of **The Very Basic Note in Combinatorics 1.5.1**. It is easier to understand this principle through examples.

Problem 1.8. Originally, 5 students are sitting in their own seats. In how many ways can they rearrange the seats such that none of them gets back to their original seats?

Solution. For simplicity, we label the students from 1 to 5. We have to count the number of permutations of $1, 2, 3, 4, 5$ such that the number 'i' does not lie in the ith place. It is hard to count directly, but it may be easier if we try to count the number of permutations where some of the numbers return to their own places.

Define $N_{i_1 \cdots i_k}$ to be the number of permutations for which $i_1, \dots, i_k$ return to their own places (other numbers may get to their own places as well). Since these k digits are fixed, we just need to permute the remaining $5 - k$ digits. Therefore, we have $N_{i_1 \cdots i_k} = (5-k)!$.

By the inclusion-exclusion principle, the number of permutations with at least one number getting back to its own place is

$$\begin{aligned}&N_1 + N_2 + \cdots + N_5 - N_{12} - N_{13} - \cdots - N_{45} + N_{123} + N_{124} + \cdots + N_{345} \\ &- N_{1234} - N_{1235} - \cdots - N_{2345} + N_{12345} \\ =& 4! \times \binom{5}{1} - 3! \times \binom{5}{2} + 2! \times \binom{5}{3} - 1! \times \binom{5}{4} + 0! \times \binom{5}{5} \\ =& 76.\end{aligned}$$

Then there are $5! - 76 = 44$ permutations such that all numbers do not lie in their own places. $\square$

Problem 1.9. Find the number of 5-digit numbers whose sum of digits is 13.

Solution. Suppose the 5-digit number is $\overline{abcde}$. The conditions are $a+b+c+d+e = 13$ with $1 \leqslant a \leqslant 9$ and $0 \leqslant b, c, d, e \leqslant 9$. Similar to problem 1.4(c), we let $a' = a - 1$ so that $0 \leqslant a' \leqslant 8$ and $a' + b + c + d + e = 12$. By problem 1.4(a), there are $\binom{16}{4}$ nonnegative integer solutions to this equation. But we need to eliminate those cases where some of the variables exceed their upper bounds.

For the case $a' \geqslant 9$, we rewrite $a'' = a' - 9$ and solve $a'' + b + c + d + e = 3$ in nonnegative integers. There are $\binom{7}{4}$ solutions. For other variables, say $b \geqslant 10$, we rewrite $b' = b - 10$ and solve $a' + b' + c + d + e = 2$ in nonnegative integers. There are $\binom{6}{4}$ solutions. Similarly, there are $\binom{6}{4}$ solutions for each of the cases $c \geqslant 10$, $d \geqslant 10$ and $e \geqslant 10$.

According to the inclusion-exclusion principle, we still need to consider those cases where at least 2 variables exceed their bounds. But this is impossible since in such cases, we have $a' + b + c + d + e \geqslant 9 + 10 > 12$. Therefore, all these cases contribute nothing.

Therefore, by the inclusion-exclusion principle, the number of 5-digit numbers with sum of digits 13 is

$$\binom{16}{4} - \binom{7}{4} - 4 \times \binom{6}{4} = 1725.$$

This solution suggests a possible way to handle similar problems as problem 1.1(b). □

1.2 Recurrence Relations

We usually need to count things in a general case in terms of a positive integer n. If we can guess the answer, we may try to prove our claim by induction. However, sometimes the answer may be too difficult to guess. A useful method is to set up recurrence relations between the cases, i.e. if we define a_n to be the number of ways corresponding to the case n, then we try to find an equation relating a_n with other terms of smaller indices like a_{n-1}, a_{n-2}. Then we can use some algebraic methods to find the general formula for a_n.

Problem 1.10. Find the number of n-digit numbers ($n \geqslant 3$) formed by digits $1, 2, 3$ only so that all digits are used and no two consecutive digits are the same.

Solution. Let a_n be the number of such n-digit numbers. Clearly, $a_3 = 3! = 6$. For $n \geqslant 4$, each n-digit number satisfying the condition is one of the following form: either the first $n - 1$ digits also satisfy the condition or not.

For the former case, there are a_{n-1} choices for the first $n-1$ digits, and 2 choices for the unit digit as it is distinct from the tens digit. This gives $2a_{n-1}$ possibilities.

For the latter case, the first $n-1$ digits violate the condition since they do not contain all 3 digits. Therefore, the digit that disappears must show up in the unit digit. There are 3 choices for the unit digit and 2 choices for the tens digit. Afterwards, note that the remaining digits are uniquely determined as they are distinct from the unit digit and the digit immediately to their right. For example, if the last two digits are 12, then the number must be $\cdots 3131312$. This gives $3 \times 2 = 6$ possibilities.

Thus, we get the formula $a_n = 2a_{n-1} + 6$ for $n \geqslant 3$. Adding

$$\begin{aligned} a_n &= 2a_{n-1} + 6, \\ 2a_{n-1} &= 2^2 a_{n-2} + 2 \times 6, \\ 2^2 a_{n-2} &= 2^3 a_{n-3} + 2^2 \times 6, \\ &\ldots, \\ 2^{n-4} a_4 &= 2^{n-3} a_3 + 2^{n-4} \times 6, \end{aligned}$$

we get $a_n = 2^{n-3}a_3 + 6 + 2 \times 6 + 2^2 \times 6 + \cdots + 2^{n-4} \times 6 = 6(2^{n-2} - 1)$. □

Problem 1.11. Find the number of ways to paint the vertices of an n-gon with 3 colours so that no two consecutive vertices have the same colour. (Some colours can be left unused.)

Solution. The problem is easy if we are going to paint the points in a row instead of a cycle. But we now require the first and the last point to have different colours too. We usually use recurrence relation to handle such problems.

Let a_n be the number of ways to paint n points in a row so that every two adjacent points have distinct colours, and the first and the last points have the same colour. Similarly, let b_n be the corresponding number of ways to paint n points in a row so that every two adjacent points have distinct colours, and the first and the last points have distinct colours. Note that we are going to find b_n in this problem, but we shall find it useful to have a_n.

For a_{n+1}, since the 1st and the $(n+1)$st points have the same colour, the 1st and the nth points must have distinct colours. So there are b_n choices for the first n points and one way to colour the last point. This gives $a_{n+1} = b_n$.

Similarly, for b_{n+1}, if the 1st and the nth points have the same colour, then there are 2 choices for the $(n+1)$st point. If the 1st and the nth points have distinct colours, then there is a unique choice for the last point. This gives $b_{n+1} = 2a_n + b_n$.

We try to get an equation involving only b_n's. Indeed, we have

$$b_{n+1} = 2a_n + b_n = 2b_{n-1} + b_n.$$

We can now use the standard method to solve this linear recurrence equation. The characteristic equation of $b_{n+1} = b_n + 2b_{n-1}$ is $t^2 - t - 2 = 0$. The roots are 2 and -1. Therefore, we have

$$b_n = A \cdot 2^n + B \cdot (-1)^n$$

for some constants A and B. By definition, we deduce $b_1 = 0$ and $b_2 = 6$. Therefore, putting $n = 1$ and $n = 2$, we solve

$$\begin{cases} 2A - B = 0, \\ 4A + B = 6. \end{cases}$$

The solution is $(A, B) = (1, 2)$. Thus, we obtain $b_n = 2^n + 2 \cdot (-1)^n$. $\square$

2 Extremal Combinatorics

Extremal combinatorics refer to a kind of problems which seek for the maximum or minimum value of a quantity such that certain things can occur. To handle such problems, note that we need to establish the bound and construct a possible case for which this bound can be attained. Usually, such a construction is obvious (which also gives a hint to the bound), while obtaining the bound is more difficult. One basic technique is given below.

Theorem 2.1. (Pigeonhole principle(抽屜原理/鴿巢原理)) Suppose there are n elements each of which belongs to exactly one of the sets $S_1, S_2, \ldots, S_m$. Then there exist $\left\lceil \frac{n}{m} \right\rceil$ of them which belong to the same set.

This is theorem 2.5 of **The Very Basic Note in Combinatorics 1.5.1**.

Problem 2.1. We pick some poker cards (without jokers) randomly. At least how many cards should be picked so that there must be a club among the chosen cards? (There are 13 cards of each of the club, diamond, heart and spade suits.)

Solution. Intuitively, the 'worst' case is to pick all the diamonds, hearts and spades. There are 39 such cards. If we pick one more card, then we must get a club.

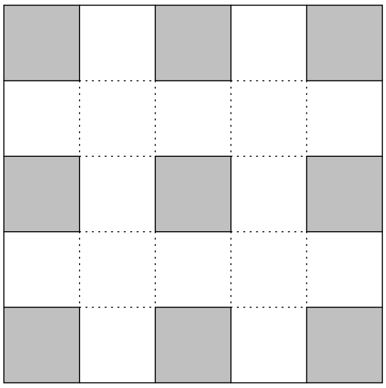
However, usually such an argument is not accepted as a formal solution. For this simple problem, the 'worst' case may be obvious so no one would challenge the above argument. But for other problems, one may find that it is difficult to define the meaning of 'worst'. It may seem to be obvious to us what a 'worst' case means, but it is not enough to convince the others. In most cases, we need to use more mathematical arguments to prove our claim.

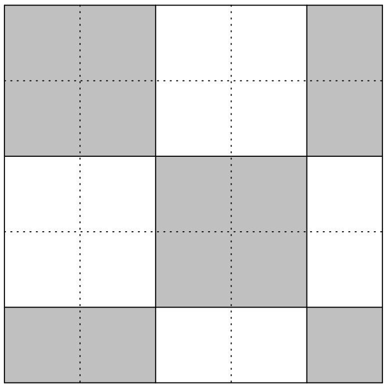
Using the above counterexample, we know that picking only 39 cards is not sufficient. We need to pick at least 40 cards. Now, assume 40 cards are picked randomly. Since there are only $13 \times 3 = 39$ cards which are not clubs, there must exist a card obtained which is a club. So picking 40 cards is sufficient. To conclude, we need to pick at least 40 cards. □

Problem 2.2. At most how many cells can be chosen from a 5×5 chessboard so that no two chosen cells are adjacent (which means they share at least one common vertex)?

Solution. It is not hard to work out the optimal case. Indeed, it is natural to start choosing a corner cell, skip a cell in the same row and choose the next cell, then skip another cell and choose another corner cell, etc. In this way, we can choose 9 cells as indicated in the left figure below.

To show that we can choose at most 9 cells, a standard way is to partition the chessboard into 9 regions such that we can choose at most 1 cell from each region. Noticing that any 2 cells in a 2×2 square are adjacent, we partition the chessboard in the way as shown in the right figure below. Every 2 cells in the same region are adjacent, and so we can choose at most 1 cell from each region. This proves at most 9 cells can be chosen. So the answer is 9.

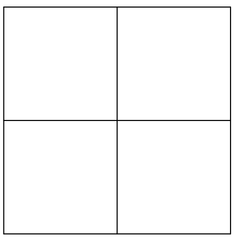




$\square$

Problem 2.3. There are n points in a square of side length 2. If the distance between every two points is larger than $\sqrt{2}$, what is the largest possible value of n?

Solution. As in the previous problem, the trick is to partition the square into some regions such that the distance between every two points in the same region is at most $\sqrt{2}$. A simple geometric meaning of $\sqrt{2}$ is the length of diagonal of a unit square. Therefore, we think of partitioning the square into 4 unit squares in the obvious way. Then the distance between any two points in the same unit square is at most $\sqrt{2}$ by Pythagoras' theorem. If $n \geqslant 5$, then two of the points must lie in the same unit square by the pigeonhole principle. This is a contradiction. Therefore, we have $n \leqslant 4$. Indeed, there can be 4 points as we can choose the 4 vertices of the square, where the distance between any two points is at least 2.



$\square$

Problem 2.4. Let S be a subset of $\{1, 2, \ldots, 100\}$ such that for any distinct $x, y \in S$, x does not divide y. Find the maximum number of elements of S.

Solution. To have $x \nmid y$ for any distinct $x, y \in S$, a simple way is to pick all large elements so that $2x > y$ whenever $x < y$. For example, we may pick $S = \{51, 52, \ldots, 100\}$. In this case, we have $|S| = 50$. Guessing the answer from this special example, we try to prove that $|S|$ cannot exceed 50.

We now partition the numbers into groups such that for any two numbers $x < y$ in the same group, we have $x \mid y$. To minimize the number of groups, we should maximize the number of elements in each group. Since a larger number is at least 2 times that of a smaller number in the same group, the 'best' strategy is to put the pairs $x, 2x$ in the same group. Indeed, we partition the numbers into the following groups:

$$\{1, 2, 4, 8, 16, 32, 64\}, \{3, 6, 12, 24, 48, 96\}, \{5, 10, 20, 40, 80\}, \ldots, \{99\}.$$

More explicitly, the first number in each group is an odd number, and all other numbers are the first number multiplied by a power of 2. There are 50 groups in total. By the pigeonhole principle, if we choose more than 50 numbers, then 2 of them belong to the same group, where one of them is a multiple of the other. This is a contradiction. Thus, the maximum value of $|S|$ is 50. $\square$

Problem 2.5. There is a permutation of $1, 2, \ldots, 10$. Suppose we can always delete k numbers in the sequence no matter how the numbers are permuted, so that the remaining numbers form a monotonic sequence (i.e. increasing or decreasing). Find the smallest possible value of k.

Solution. This time the 'worst' case is not obvious. The most special cases seem to be the natural order $(1, 2, \ldots, 10)$ or the reverse order $(10, 9, \ldots, 1)$. But both turn out to be the 'best' cases as we do not need to delete any number at all. If one tries to work on this problem, one can only try to write down some sequences really in a random way (like $4, 8, 2, 5, 7, 1, 10, 6, 3, 9$) and test for the possibly minimum number. But one probably cannot tell when one has got the 'worst' case.

Therefore, this time we shall try to obtain the bound before we guess the answer. Again a usual way is to apply the pigeonhole principle. The problem is how we can construct the pigeonholes. As this is also a standard example, its solution is well-known to most MO people. So it is difficult to tell how to think of the solution. One may try to gain more ideas and experiences through this standard solution.

For each number n in the sequence, define I_n to be the maximum number of terms of an increasing sequence using n as the initial term, and define D_n analogously to be the maximum number of terms of a decreasing sequence using n as the initial term. For example, in the sequence $4, 8, 2, 5, 7, 1, 10, 6, 3, 9$, we have $I_4 = 4$ and $D_4 = 3$ (corresponding to the subsequences $4, 5, 7, 10$ and $4, 2, 1$), etc.

We first prove that for any $m \neq n$, either $I_m \neq I_n$ or $D_m \neq D_n$. Suppose on the contrary that $I_m = I_n$ and $D_m = D_n$ for some $m < n$. If m appears before n, we first find an increasing subsequence $a_1, a_2, \ldots, a_{I_n}$ with $a_1 = n$. Such a subsequence exists by the definition of I_n. Since $m < n$ and m appears before n, the sequence $m, a_1, a_2, \ldots, a_{I_n}$ is also increasing. However, it has $I_n + 1 = I_m + 1$ terms, which contradicts the maximality of I_m. Similarly, if m appears after n, then the condition $D_m = D_n$ is violated by adding n before the longest decreasing subsequence beginning from m.

We are now able to get a bound using the above observation. We claim that there exists some I_n or D_n which is at least 4. Suppose all these values are at most 3. Note that they must be positive integers. So there are $3 \times 3 = 9$ choices for the pair (I_n, D_n). As there are 10 numbers, 2 of them correspond to the same pair $(I_m, D_m) = (I_n, D_n)$ by the pigeonhole principle. This contradicts our observation. Thus, the claim is verified. Therefore, by leaving such a sequence, we can always get a monotonic sequence by deleting $10 - 4 = 6$ numbers.

It now suffices to provide a case such that we need to delete at least 6 numbers. There should be many such sequences. But in order to explain easily and to generalize the proof, we may need a more systematic construction. For example, we may consider $3, 6, 9, 2, 5, 8, 1, 4, 7, 10$. For increasing sequences, we can only choose 1 from each of $\{3, 2, 1\}, \{6, 5, 4\}, \{9, 8, 7\}, \{10\}$. For decreasing sequences, we can only choose 1 from each of $\{3, 6, 9\}, \{2, 5, 8\}, \{1, 4, 7, 10\}$. So no 5 terms can form a monotonic sequence.

In general, the Erdős-Szekeres theorem states that any sequence of real numbers having at least $(m-1)(n-1)+1$ terms must contain a monotonic increasing subsequence of length m or a monotonic decreasing subsequence of length n. $\square$

3 Mathematical Games

Mathematical games usually refer to some games played between two players. The players need to follow the rules and look for a strategy to win the game. Very often we have to determine which player has the winning strategy. Since the rules depend on the game, it is not guaranteed that the player who moves first can always win. Also, the greedy algorithm does not always work. This means the 'best' choice at each step may not give the best result on the whole.

The only general strategy which always applies is to try to play the game to get familiar with the rules. Also, we may start from some smaller cases and look for patterns to see what is the key to win the game.

Problem 3.1. (Bachet's game(巴什博弈))

(a) There are 10 coins on the table. Starting from X, the players X and Y alternately take away some coins from the table. Each time a player can take away 1, 2 or 3 coins. The one who takes away the last coin will win. Who has a winning strategy?

(b) There are 20 coins on the table. Starting from X, the players X and Y alternately take away some coins from the table. Each time a player can take away 1, 2 or 4 coins (but not 3 even if there are only 3 coins left). The one who takes away the last coin will win. Who has a winning strategy?

Solution.

(a) Although one of the players, say X, cannot control how the opponent plays, X may be able to control the outcome in each round. For example, whenever Y takes away 1, 2 or 3 coins, then X can take away 3, 2 or 1 coin correspondingly. In total, 4 coins are taken away in each round.

This observation is the only trick to play this game. X can first take away 2 coins, leaving 8 coins on the table. Using the above strategy, 4 coins will be taken away in the next round (Y's and X's turns), and 4 more coins in the further round. In particular, the last coin must be taken away by X so that X can win.

In general, if there are $4k$ coins ($k \in \mathbb{N}$) on the table initially, then Y has a winning strategy. Otherwise, X can always win. $\square$

(b) This time we cannot use the same strategy as in part (a). For example, if we want to take away 5 coins in each round, then we have no choice when the opponent takes away 2 coins. Indeed, by noting that no one can take away 3 coins (or multiples of 3) at a time, we consider modulo 3.

X can win by taking 2 coins initially, leaving 18 coins. After Y's move, the number of coins is congruent to 1 or 2 modulo 3. Then X can take away 1 or 2 coins accordingly so that the remaining number of coins is divisible by 3. In this way, there must be some coins left after Y's turn since the number of coins is not congruent to 0 modulo 3. After finitely many steps, there must be 0 coins remaining, and it must be X who wins the game. □

In these examples, we have demonstrated the use of invariance. This means we may try to find some quantities or properties which remain unchanged after each step. For example, in part (a), the one who can make the number of coins become a multiple of 4 has a winning strategy since the opponent cannot keep this property, and the game ends when there is no coin (where 0 is also a multiple of 4).

Problem 3.2. A king is placed somewhere on an 8×8 chessboard. Starting from X, the players X and Y alternately move the king to an adjacent cell (a cell sharing a common vertex). Suppose the king is not allowed to return to the same cell and the one who cannot move will lose. Who has a winning strategy?

Solution. In general, the king can move in one of the 8 possible directions. So it is hard to guess how the opponent plays. We try to work out a plan so that we know how to move no matter how the opponent moves.

Firstly, we partition the 64 cells into 32 groups of 1×2 pieces. For example, we can partition each row into 4 such pieces. Regardless of the initial position of the king, X can move the king from the cell it was to the cell within the same pair. Afterwards, Y must move the king to a new pair in each step, and X can move the king to the other cell of the same pair accordingly. Since the cells in the same pair are adjacent cells, and the other cell has not been travelled, X can always make a valid move. For Y's turn, both cells of each pair have been travelled or have not been travelled. That means Y must move the king to a new pair of cells. Then X can continue to use the strategy.

As there are only 64 cells, Y will lose eventually. One can also work out a winning strategy for an $m \times n$ board in general. □

Problem 3.3. There is a round table of radius 10, and there are sufficiently many circular coins of radius 1. Starting from X, the players X and Y alternately put a coin on the table such that the coin does not lie outside the table, and no two coins overlap. The one who cannot move will lose. Who has a winning strategy?

Solution. This is another well-known mathematical game. At first glance it may look difficult since there are infinitely many ways to put the coins. The key idea in solving this problem is symmetry, which is another common technique.

We show that X has a winning strategy. The idea is to put the first coin at the centre so that the table and this coin are concentric. Afterwards, whenever Y puts a coin somewhere, X always puts a coin in the position symmetric to that of Y's coin about the centre of the table. Since the table is symmetric about the centre, X can always do this without violating the rules, as long as Y does not. As only finitely many coins can be put on the table, the game must eventually end, and it must be X who wins the game. $\square$

Problem 3.4. There is an 8×8 chessboard. Starting from X, the players X and Y alternately put white and black chess pieces on some unoccupied cells. A chess piece cannot be put on the cell which lies on the same diagonal line (a line making an angle of $45°$ with the boundary of the chessboard) as a chess piece of different colour. The one who cannot move will lose. Who has a winning strategy?

Solution. This is another game using the symmetry technique. However, we cannot use the symmetry about the centre, since there are some cells symmetric about the centre which lie on the same diagonal line. Rather than that, we use the symmetry about the horizontal midline ℓ of the chessboard. Indeed, whenever X puts a chess piece somewhere, Y always puts a chess piece on the cell symmetric to that of X's chess piece about ℓ. Since these chess pieces lie on the same vertical line, they do not lie on the same diagonal line. Also, whenever X's move does not violate the rule, so does Y's move by symmetry. This shows Y can always make a valid move. After finitely many moves, X must eventually lose, and hence Y has a winning strategy. $\square$

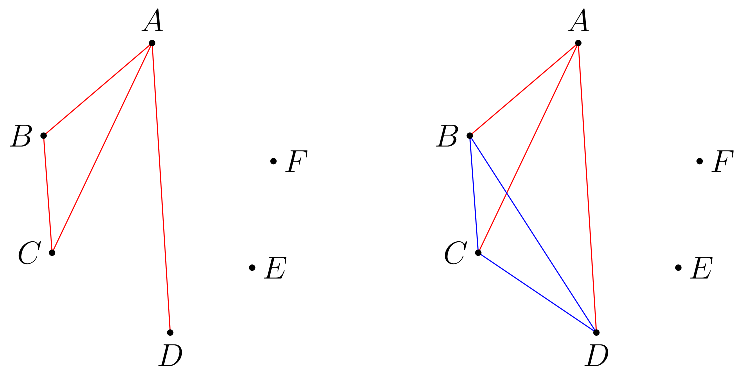
4 Graph Theory

Some combinatorial problems can be rephrased as problems in graph theory. When solving such problems, the tools are mostly the same as other combinatorial problems. However, sometimes we can use some well-known results in graph theory to simplify the proof. In this basic note, we are going to solve several well-known results.

Problem 4.1. Let G be a complete graph of 6 vertices. We colour each edge in red or blue. Prove that there is a monochromatic triangle in this graph.

Solution. This is a basic result in the Ramsey theory(拉姆齊理論). The key idea is to use the pigeonhole principle.

Pick any vertex A. Among the 5 edges incident on A, at least $\left\lceil \frac{5}{2} \right\rceil = 3$ of them have the same colour by the pigeonhole principle. Without loss of generality, assume AB, AC, AD are red. Among the edges BC, CD, DB, if one of them is red, say BC, then $\triangle ABC$ is a red triangle. Otherwise, all of BC, CD, DB are blue, and so $\triangle BCD$ is blue. This proves the assertion.



$\square$

Problem 4.2.

(a) Let G be a simple graph of 7 vertices and 5 edges. Prove that the graph is disconnected.

(b) Let G be a simple connected graph of 7 vertices and 6 edges. Prove that there is no cycle in this graph.

(c) Let G be a simple connected graph of 7 vertices and 7 edges. Prove that there exists a cycle in this graph.

Solution.

(a) We can prove by induction that any connected graph of n vertices contains at least $n-1$ edges. This clearly proves the result.

For $n = 2$, since the only 2 vertices are connected, there is at least 1 edge.

Assume the assertion holds for some $n = k \geqslant 2$. Suppose on the contrary that there is a connected graph of $k+1$ vertices having at most $k-1$ edges. By the handshaking theorem, the sum of degrees of all vertices is at most $2(k-1)$. By the pigeonhole principle, there exists a vertex A whose degree is at most $\left\lceil \frac{2(k-1)}{k} \right\rceil < 2$. Since the graph is connected, we must have $\deg A = 1$.

Now, we remove A together with the edge incident on A. Note that the remaining graph has k vertices and it is still connected. Indeed, if there are at least 2 components which are not connected, then A must be joined to at least one vertex in each component, contradicting $\deg A = 1$. Therefore, by the inductive hypothesis, there are at least $k-1$ edges remaining. Together with the edge removed, there are at least k edges in the original graph, contradiction. This proves the inductive step, and we are done. □

(b) Suppose on the contrary that there exists a cycle. If we remove one edge from the cycle, the graph is still connected (because if we need to go from one vertex to another vertex via the deleted edge, we can go around the cycle instead). This gives us a simple connected graph of 7 vertices and 5 edges, contradicting part (a). Therefore, the original graph has no cycle.

In general, a simple connected graph of n vertices and $n-1$ edges has no cycle. Note that the connectedness is necessary. □

(c) We use the same idea as in part (a). If there exists a vertex of degree 1, we remove this vertex together with the edge incident on it. Each time we remove exactly 1 vertex and 1 edge, and the remaining graph is still connected. Therefore, by repeating the process, we eventually obtain a connected subgraph H of k vertices and k edges such that the degree of each vertex is at least 2. (Note that $k \geqslant 3$, so it is impossible that all vertices are removed.)

Since the total degree of the vertices in H is $2k$ and each of the k vertices has degree at least 2, the degree of each vertex must be 2. Starting from an arbitrary edge A_1, A_2, we can add more and more vertices to form a longer path as follows. Since $\deg A_2 \geqslant 2$, we can form a longer path A_1, A_2, A_3. Similarly, since $\deg A_3 \geqslant 2$, A_3 is joined to another vertex A_4 (possibly A_1). Repeating the process, we must eventually obtain a path $A_1, A_2, \ldots, A_m$ such that A_m is adjacent to one of $A_1, A_2, \ldots, A_{m-2}$. Note that A_m cannot be adjacent to $A_2, A_3, \ldots, A_{m-2}$ since the degrees of these vertices are already 2. This shows A_m is adjacent to A_1, and so we obtain a cycle $A_1, A_2, \ldots, A_m, A_1$. This must be a cycle in the original graph as well. (In fact, note that we must have $m = k$ since otherwise the other vertices cannot be connected to $A_1, A_2, \ldots, A_m$.)

In general, any simple graph of n vertices and at least n edges must contain a cycle. $\square$

Problem 4.3.

(a) Prove that a finite simple connected graph has an *Eulerian circuit*(歐拉迴路) (a circuit that visits every edge exactly once) if and only if the degree of every vertex is even.

(b) Prove that a finite simple connected graph has an *Eulerian trail*(歐拉路徑) (a walk that visits every edge exactly once) if and only if the number of vertices having odd degree is 0 or 2.

Solution. These results tell us whether we can draw a connected graph with a pencil so that the pencil is always in contact with the paper.

(a) Suppose an Eulerian circuit exists. Suppose we move along the circuit from any starting point. Every time when we visit a vertex, we must leave that vertex using another edge. So we can pair up the edges incident on a particular vertex A (if A is the starting point, the first edge of the circuit is paired up with the last edge). Since all edges are visited exactly once, this shows the degree of A is even, and this holds for any vertex A.

Conversely, suppose the degree of every vertex is even. We start from an arbitrary vertex A_1 and move along any unused edges until this cannot be done. We obtain a walk $A_1, A_2, \ldots, A_k$ in this way that visits different edges. Note that we have used an even number of edges incident on the vertices in this walk, possibly except A_1 and A_k (which happens when $A_1 \neq A_k$). Since $\deg A_k$ is even and we cannot find another unused edge incident on A_k, it must be the case that $A_k = A_1$. In other words, we have obtained a circuit.

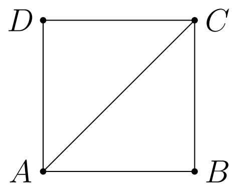
If the circuit contains all edges, we are already done. Otherwise, there must be an edge incident on some vertex, say A_1, which is unused. We repeat the same process and construct another walk starting from A_1 that visits different edges (and different from those used in the above circuit). Suppose we obtain another walk $A_1, B_2, B_3, \ldots, B_\ell$. As above, we must have $B_\ell = A_1$. Then we can combine the two circuits to form a larger circuit $A_1, A_2, \ldots, A_{k-1}, A_1, B_2, B_3, \ldots, B_{\ell-1}, A_1$. Again, repeating the same process, we eventually obtain an Eulerian circuit. This completes the proof. $\square$

(b) Suppose an Eulerian trail exists. As in the proof of part (a), we can pair up the edges incident on a vertex possibly except the first edge and the last edge of the Eulerian trail. If the starting point coincides with the terminal point, the Eulerian trail is just an Eulerian circuit, and so all vertices have even degree. If they are not the same vertex, then only the starting point and the terminal point have odd degrees, and all other vertices have even degrees.

Conversely, the case when there is no vertex having odd degree is the case in part (a), so we only consider the case when there are exactly 2 vertices A and B having odd degree. Since the graph is connected, we can find a path from A to B. If this path is removed, the degrees of all vertices will become even. Thus, we can use the same construction as in part (a) to extend this path to a walk that visits every edge exactly once. (For example, if $A, C_1, \ldots, C_k, \ldots, C_\ell, B$ is a walk, and we form another circuit $C_k, D_1, \ldots, D_m, C_k$, then we can extend the walk to $A, C_1, \ldots, C_k, D_1, \ldots, D_m, C_k, \ldots, C_\ell, B$.) □

Problem 4.4. Let G be a simple graph of 10 vertices and 26 edges. Prove that there exist 3 vertices forming a cycle.

Solution. In general, we shall prove that for a simple graph of $2n$ vertices and at least $n^2 + 1$ edges, there exist 3 vertices forming a cycle. This can be proved by induction.



For $n = 1$, there cannot be 2 edges in a simple graph of 2 vertices. For $n = 2$, if there are at least 5 edges among 4 vertices, then we must have the above graph (where BD may be an edge or not). Clearly, a triangle does exist, e.g. A, B, C, A. The base cases are done. We now assume the claim is true for $n = k$ and work on the case $n = k + 1$.

Suppose there are $2k+2$ vertices and at least $(k+1)^2+1 = k^2+2k+2$ edges. Take any edge AB. Let $C_1, C_2, \ldots, C_{2k}$ be the other vertices. If any C_j is adjacent to both A and B, then A, B, C_j, A is a triangle. Otherwise, there are at most $2k$ edges between one of A, B and one of the C_j's. Thus, there are at least $(k^2+2k+2)-1-2k = k^2+1$ edges among the C_j's. By the inductive hypothesis, we can find a triangle among the C_j's. Therefore, the claim is true for $n = k+1$. By induction, this is true for any positive integer n. □